SECURITY



Advancing Data Center Trust using TCG Technologies

Open Standards for Device Identity, Attestation, and Post-Quantum Readiness



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Trusted Computing Group

- TCG is a not-for-profit industry standards organization founded in 2003
- Open, vendor-neutral specifications supporting hardware-based Roots of Trust
- Billions of devices already incorporate TCG technologies

Collaborative work directly supports OCP Security Project priorities: hardware-rooted attestation, secure component identity, and post-quantum readiness

Promoters

AMD Promoter	Arm Promoter	Cisco Promoter	Dell Promoter	Google Promoter
HP Inc Promoter	HPE Promoter	Huawei Promoter	IBM Promoter	Infineon Promoter
Intel Promoter	Lenovo Promoter	Microsoft Promoter	NVIDIA Promoter	Toyota Promoter

Contributors

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Seagate Contributor	SEALSQ Contributor	Silicon Motion Contributor	Solidigm / SK hynix Contributor	STMicroelectronics Contributor	Texas Instruments Contributor	Toshiba Contributor	ULINK Contributor	Western Digital Contributor	Winbond Contributor

Adopter

Absolute Software Adopter	CS Services Adopter	Fagundez Adopter	JP Sa Couto Adopter	Leonardo DRS Adopter
Login Informatica Adopter	Montage Adopter	ScaleFlux Adopter	SecEdge Adopter	Supermicro Adopter
Ubiquitous Adopter	Wibu-Systems Adopter	WinMagic Adopter		

Product categories

Server	Client	Data centers
Networking	Automotive	Aerospace

…

Core Technologies

Trusted Platform Module

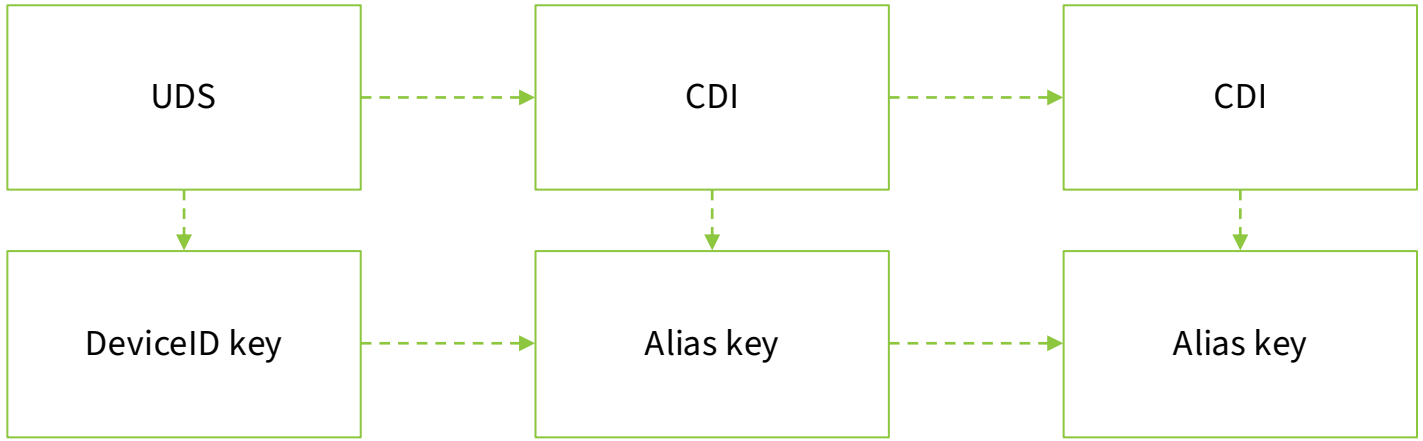
Device Identity Composition Engine

Storage device APIs

More than 90 TCG specifications and guidance documents published

Device Identifier Composition Engine (DICE)

- Lightweight hardware Root of Trust
- Establishes strong cryptographic identity and enables attestation



DICE Protection Environment (DPE)

- CDI values and private keys never leave the DPE
- Enables safe use of DICE identities by other entities in the system
- DICE identity, measured boot, and attestation services to the rest of the SoC
- Signing oracle and supports authentication to discrete TPMs
- Useful for TPM-over-SPDM, to address active-interposer mitigations

Defeating Active Interposers

Previously at OCP:

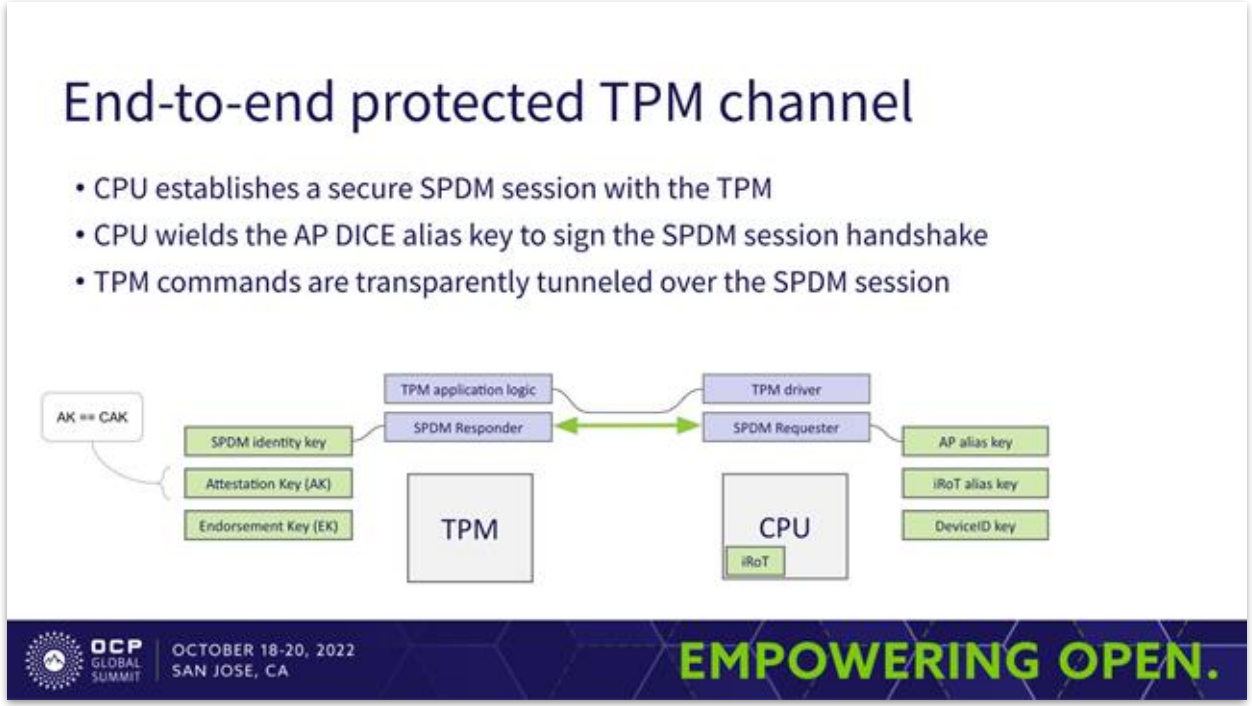
- High-level architecture for mitigating advanced TPM interposer attacks



Defeating Active Interposers

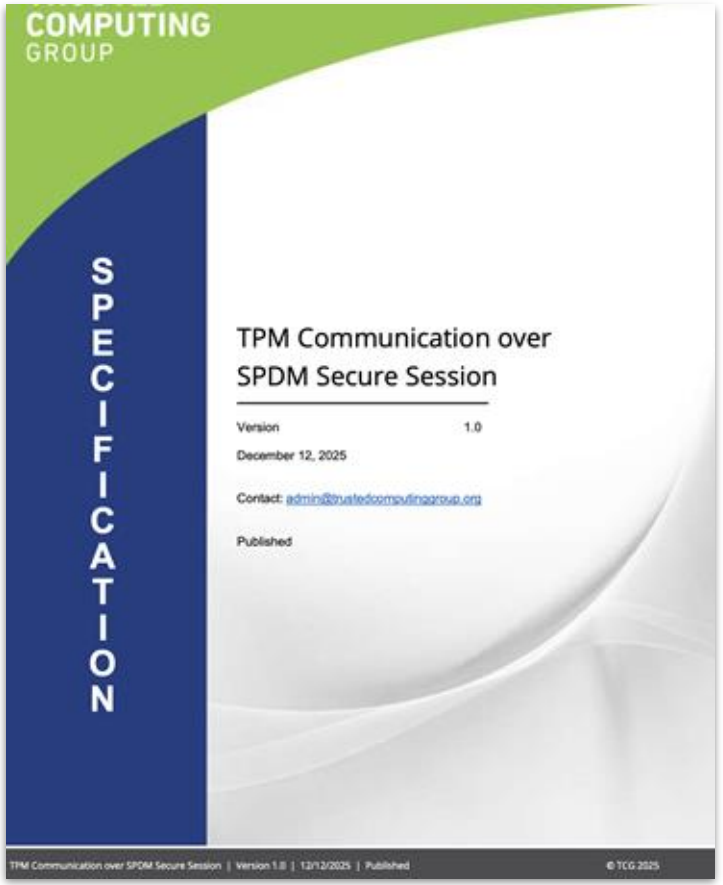
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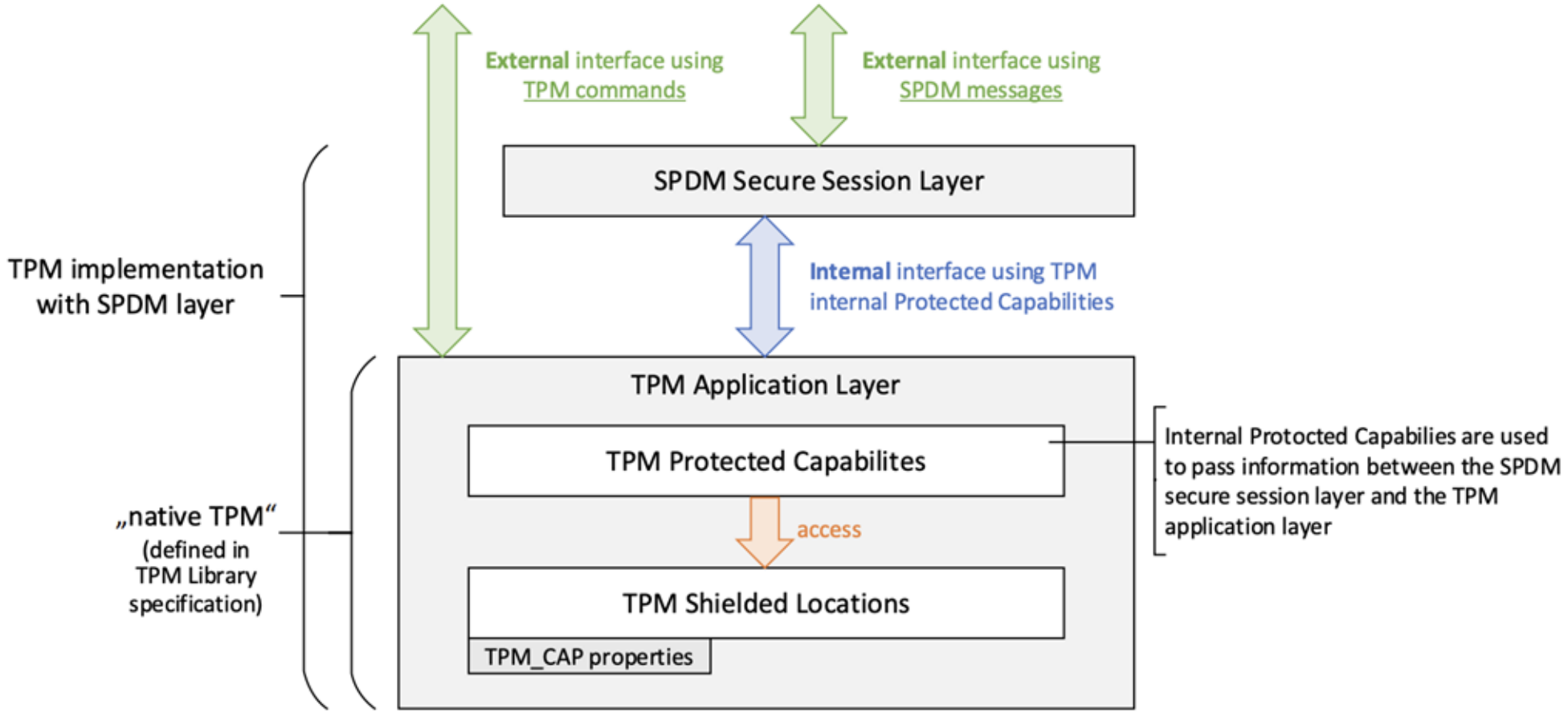
TPM-over-SPDM

- TCG Server Workgroup released a specification: TPM Communication over SPDM Secure Session[1]
- Scope: Server and Data Center
- Goal: establish a secure channel between a host (acting as Requester) and a TPM (acting as Responder)

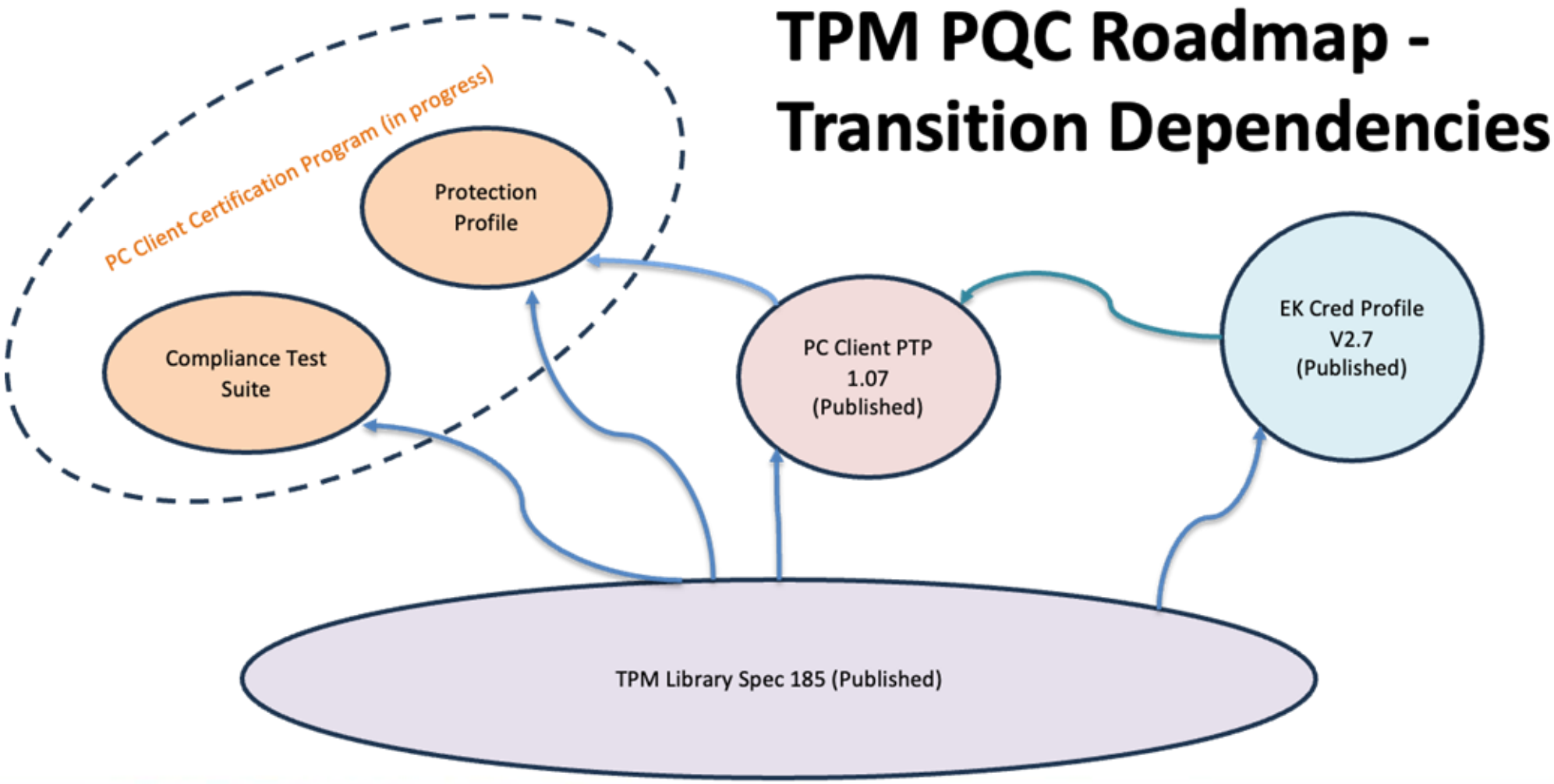


[1] https://www.trustedcomputinggroup.org/resources/documents/tpm-communication-over-spdm-secure-session

TPM-over-SPDM



Post-Quantum TPM



Call to Action

- Adopt TCG open standards to build verifiable trust chains from manufacturing through deployment and operation:
  - DICE for component-level Roots of Trust and identity
  - TPM over SPDM for end-to-end protected TPM channels
  - PQC-ready TPMs (Library Spec 1.85 + PC Client PTP 1.07)
- Engage in the TCG Server, DICE, and related work groups and in OCP Security

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@TrustedComputin

Thank You!

